## Inverse functions

1 Graph $y = x$ on the same axes as the graphs of the two functions and look to see one of the graphs is a reflection of the other about $y = x$. If they are reflections, then they are inverses.

2

a No

b Yes

c No

d Yes

e No

f Yes

g No

3

a $y = \dfrac{x+8}{7}$

b $y = \dfrac{3}{4}(x+5)$

c $y = \sqrt[3]{\dfrac{x}{6}}$

d $y = x^3 + 2$

e $y = \dfrac{8}{x-5} + 9$

f $y = \sqrt[3]{x-3} + 4$

g $y = \dfrac{8}{5x} - \dfrac{3}{5}$

h $y = \sqrt[3]{\dfrac{7}{x+3}} + 8$

i $y = \dfrac{\sqrt[3]{\dfrac{x+7}{4}} + 2}{9}$

4

a

i $f^{-1}(x) = \dfrac{1}{8}(x+9)$

ii $[-41, 7]$

iii $[-4, 2]$

b

i $f^{-1}(x) = \sqrt{x}$

ii $[0, \infty)$

iii $[0, \infty)$

c

i $f^{-1}(x) = \sqrt{16 - x^2}$

ii $[0, 4]$

iii $[0, 4]$

5

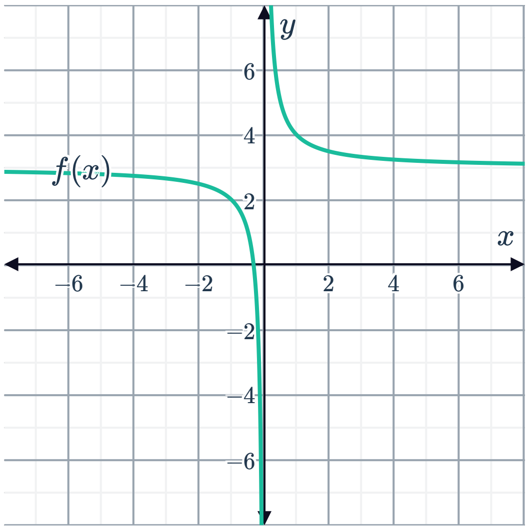
a $y = -\dfrac{4}{x}$

b $y = (x-5)^3$

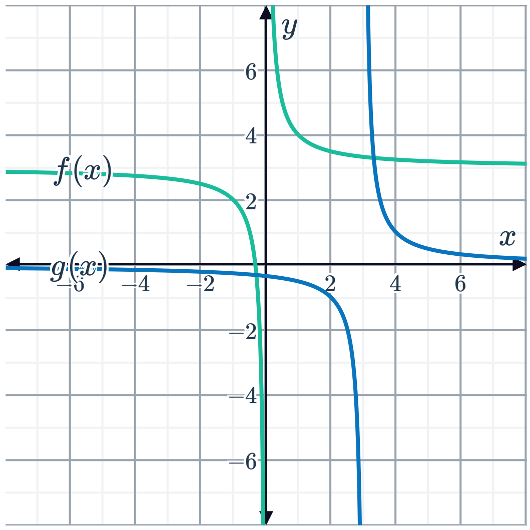
## Graphs of inverse functions

**6**

**a**

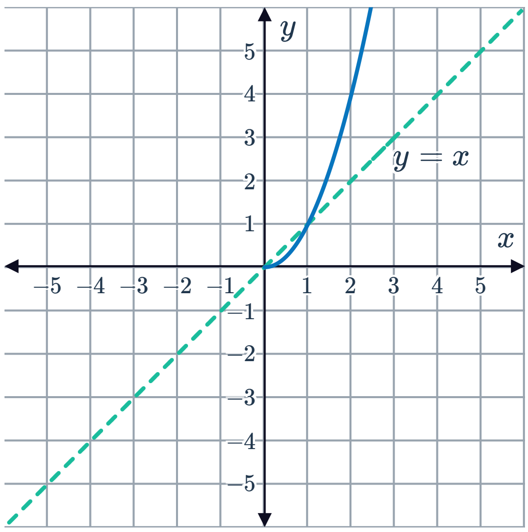



**b**




**c** Yes

**7**




**8**

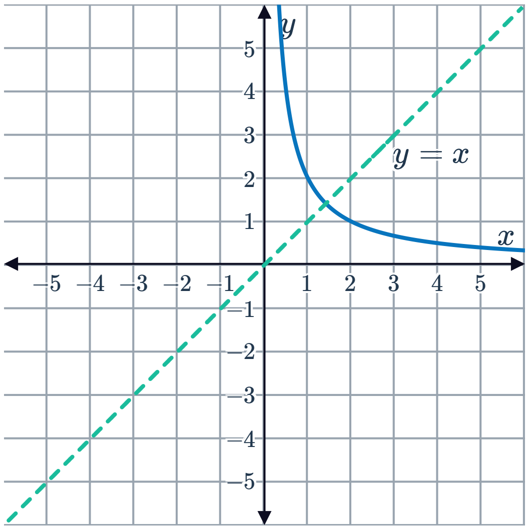
**a** $x^3 - 2$ **b** $\sqrt[3]{x+2}$ **c** $x$ **d** $x$

**e**

**i** Yes **ii** No **iii** Yes **iv** Yes

**9**

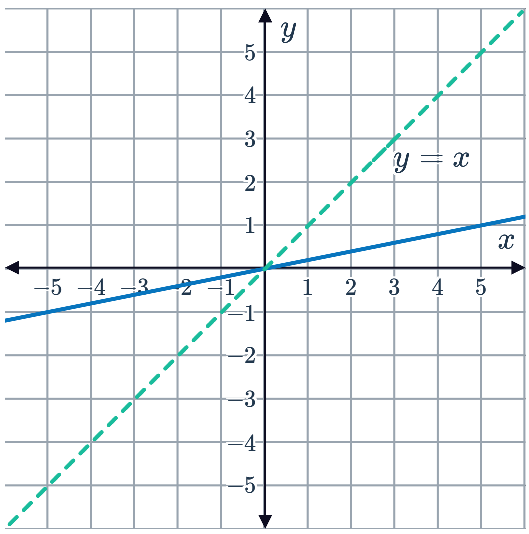
**a**



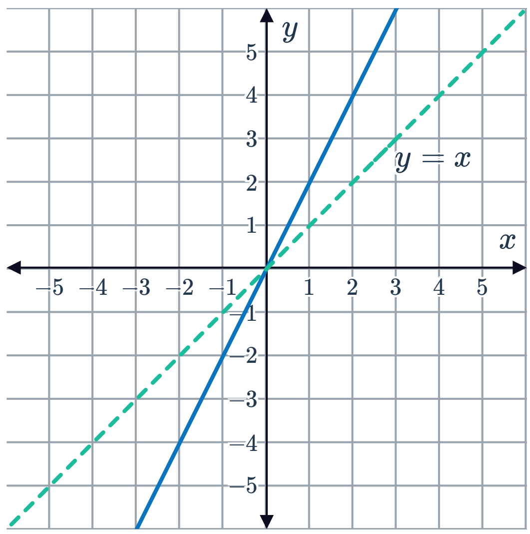

**b** The inverse graph is exactly the same as the original graph.

**10**

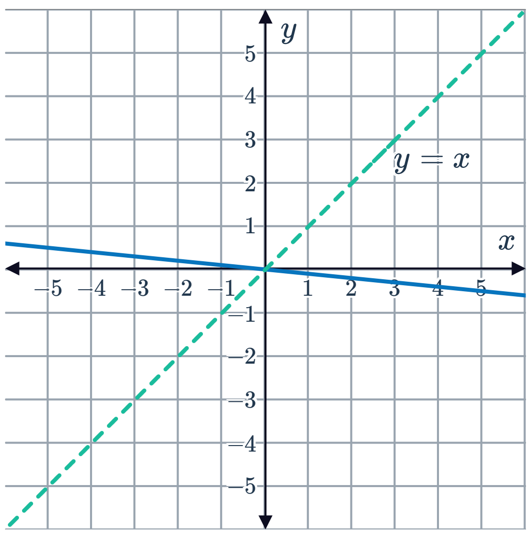
**a**



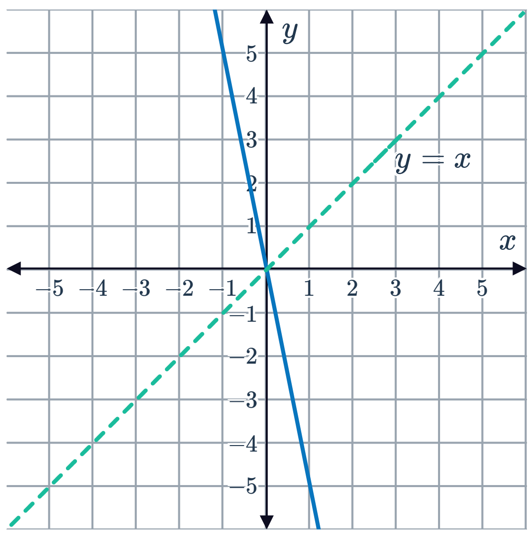

**b**



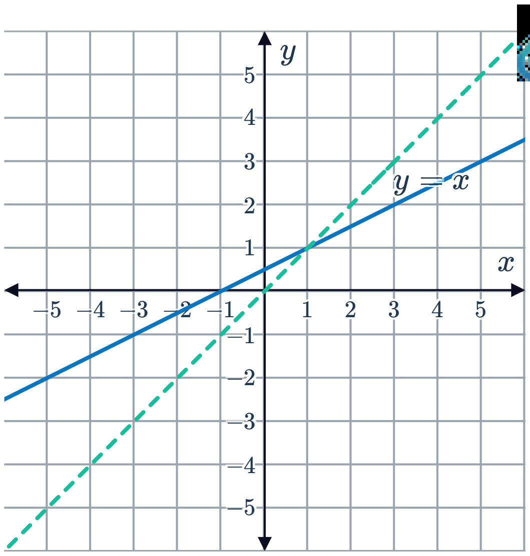

**c**



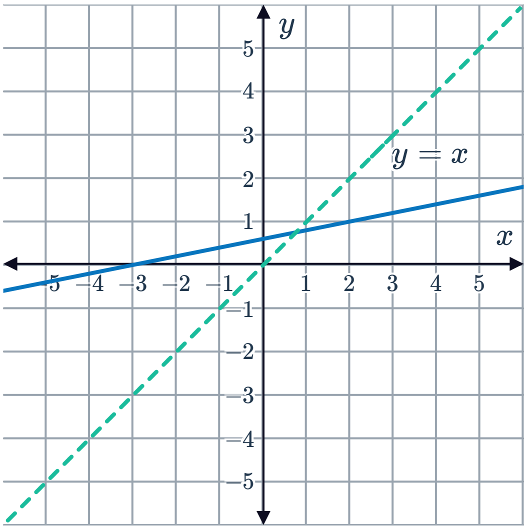

**d**



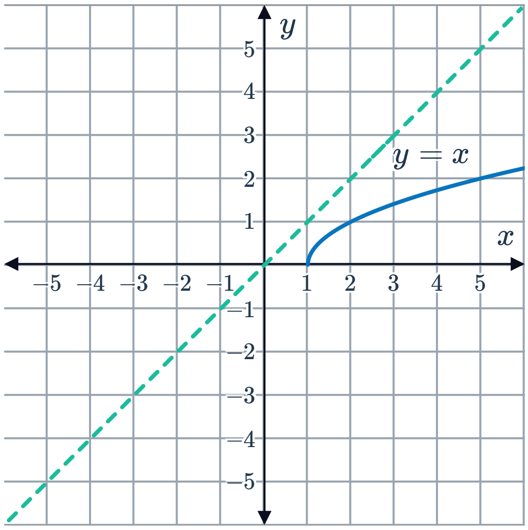
e



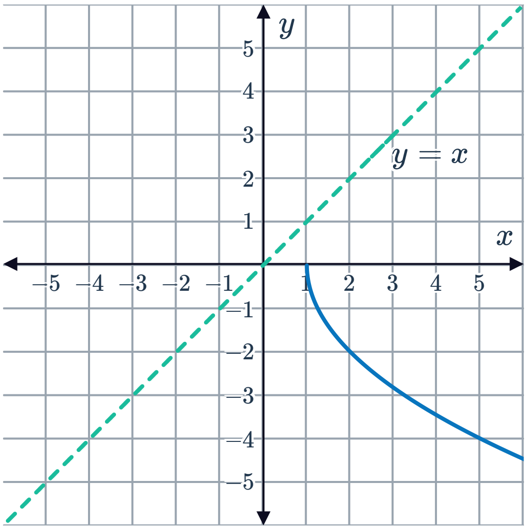

f



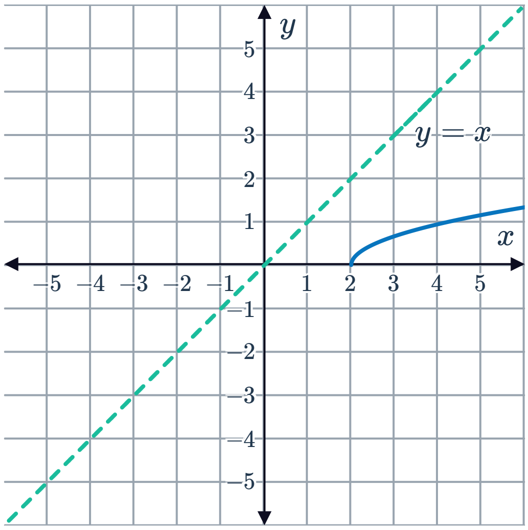

g



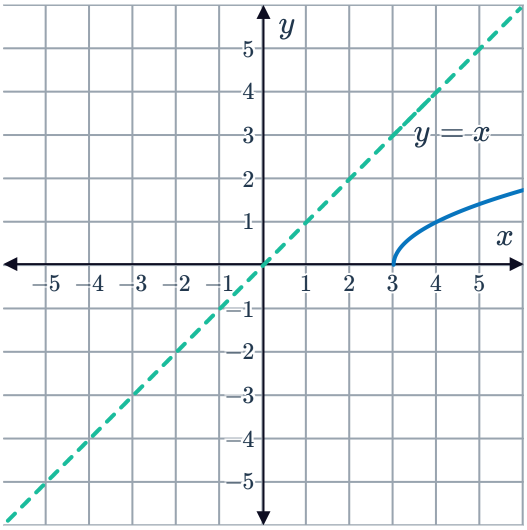

h




i



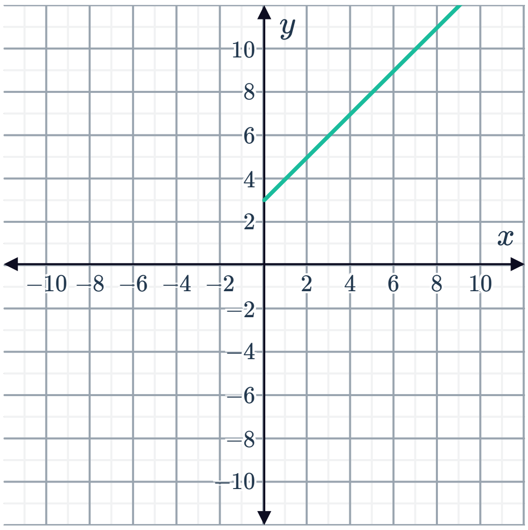

j



**11**

**a**

**i**

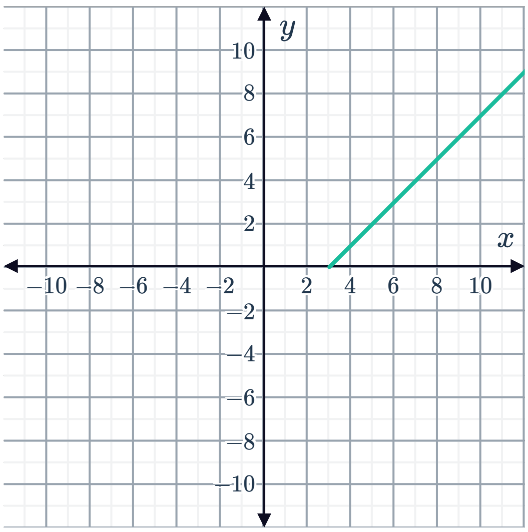



**ii** $f^{-1}(x) = x - 3$

**iii** $[3, \infty)$

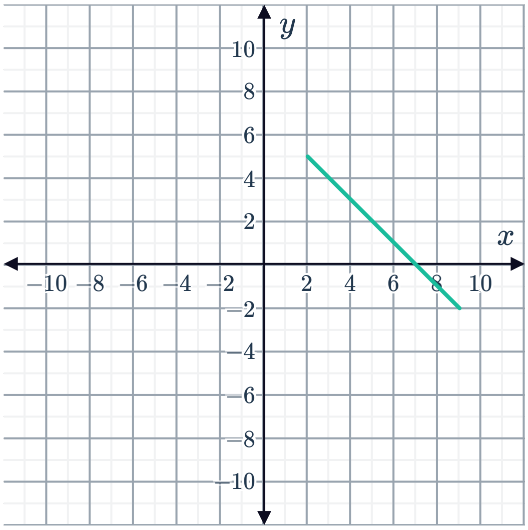
**iv** $[0, \infty)$

**v**



**b**

**i**

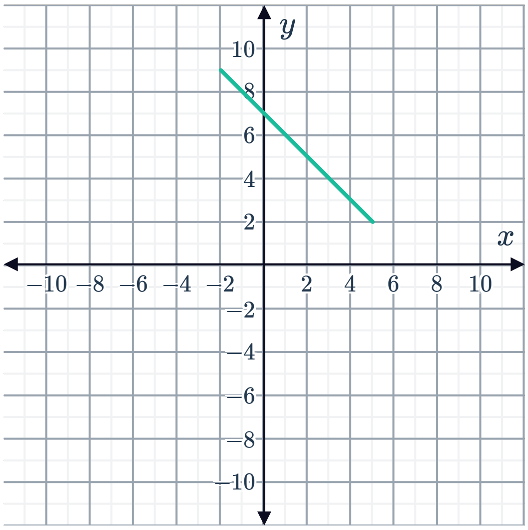



**ii** $f^{-1}(x) = 7 - x$

**iii** $[-2, 5]$

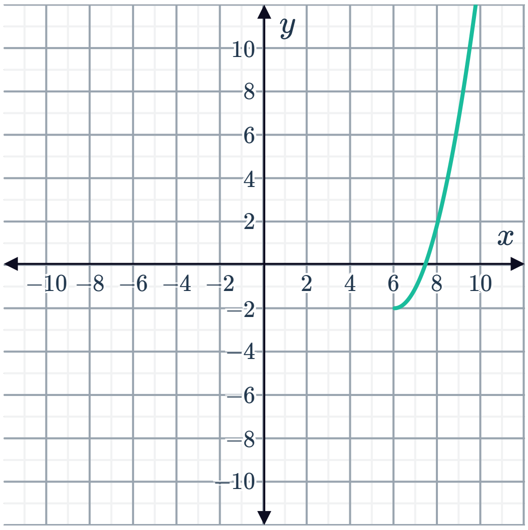
**iv** $[2, 9]$

**v**



**c**

**i**

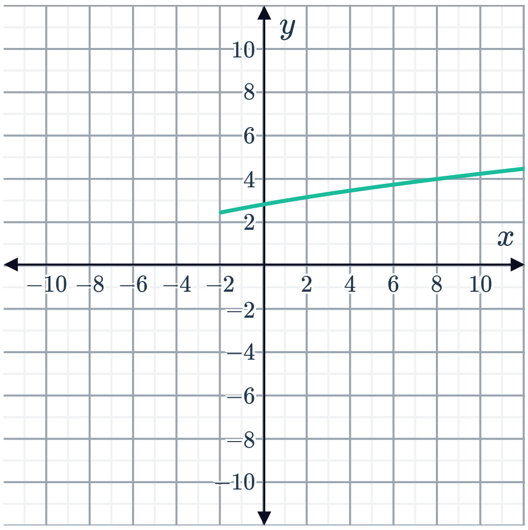



**ii** $f^{-1}(x)=\sqrt{x+2}+6$

**iii** $[-2,\infty)$

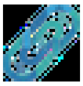
**iv** $[6,\infty)$

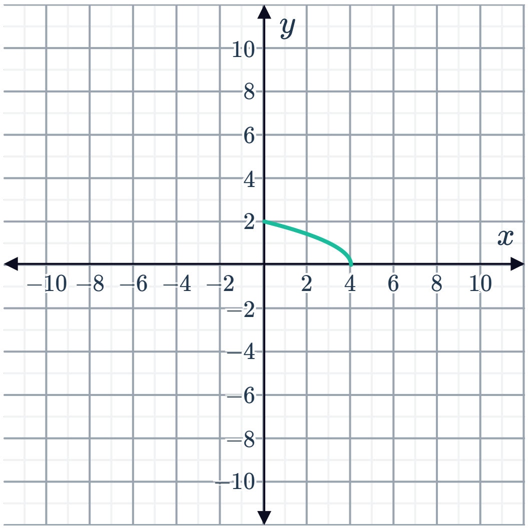
**v**



d

i



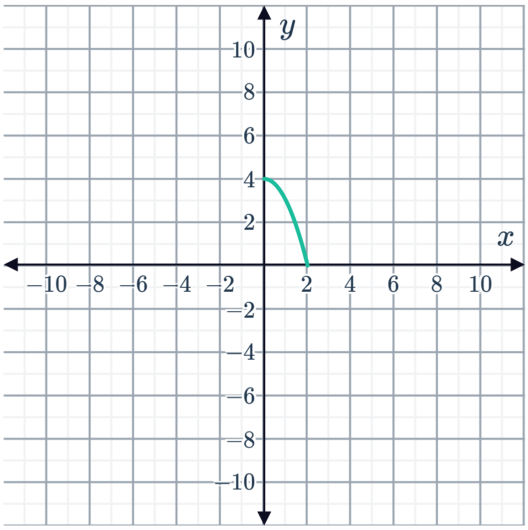



ii $f^{-1}(x) = 4 - x^2$

iii $(0, 2]$

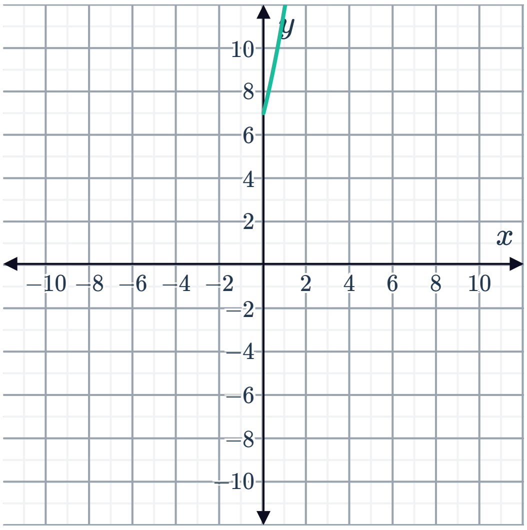
iv $[0, 4)$

v



e

i

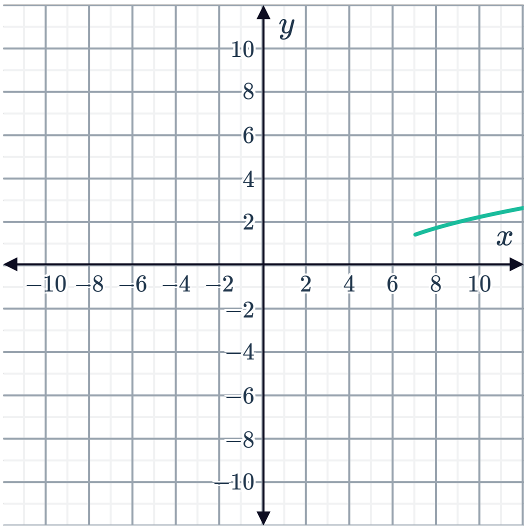



ii $f^{-1}(x) = \sqrt{x-3} - 2$

iii $[7, \infty)$

iv $[0, \infty)$

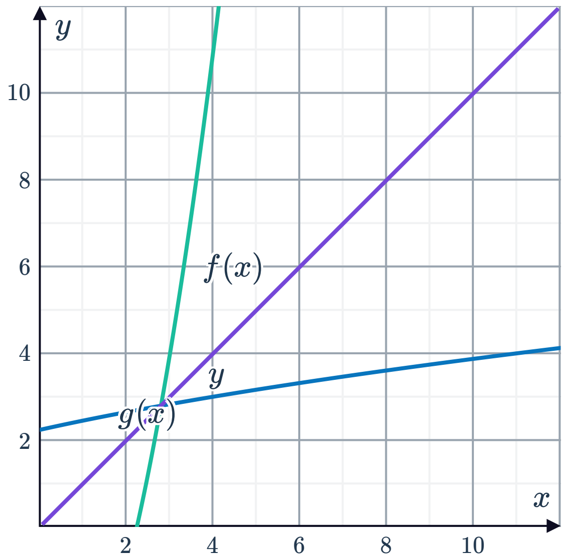
v




**12**

a $y = x$

b




c $f(x)$ is a reflection of $g(x)$ about $y = x$.

## Applications

**13**

a $d = 4.9t^2$

b $d = 122.5$

**14**

a $-25^{\circ}$

b $30^{\circ}$

c $T^{-1}(x) = \frac{9}{5}x + 32$

d For converting Celsius temperatures $x$ to Fahrenheit temperatures $T^{-1}(x)$

**15**

a $t = \frac{5}{6}$

b $t = \frac{80 - \sqrt{v}}{96}$

**16**

a $T = 1.07A$

b $A = \frac{T}{1.07}$

**17**

a $l = \frac{9.8T^2}{4\pi^2}$

b $l = 0.6 \text{ m}$